



2 Numbers are stored in a computer using binary floating-point representation with:

- 12 bits for the mantissa
- 4 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the normalised binary floating-point representation of +124.4375 in this system. Show your working.

Mantissa

Exponent

--	--	--	--	--	--	--	--	--	--	--	--

--	--	--	--

Working

.....

.....

.....

.....

.....

[3]

(b) Calculate the denary value of the following normalised binary floating-point number. Show your working.

Mantissa

Exponent

1	0	1	0	0	0	1	0	1	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---

0	1	1	0
---	---	---	---

Working

.....

.....

.....

.....

Denary value

[3]

